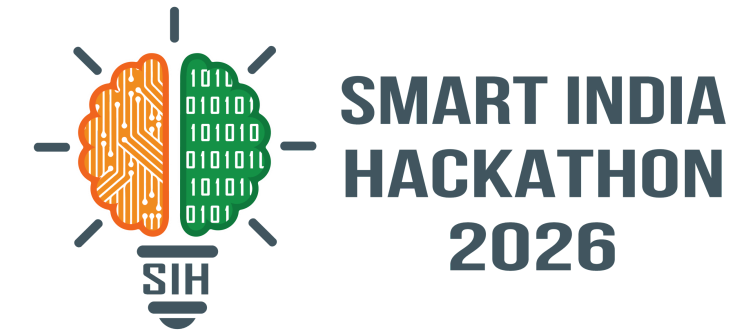


SMART INDIA HACKATHON 2026



- Problem Statement ID - 26017
- Problem Statement Title - Predictive Analytics
System for Early Detection of Land Acquisition
Delays
- Theme - Smart Automation
- PS Category - Software
- Team ID - 152
- Team Name - Navotkarsh



LandSight AI - Enabling Timely Land, Enabling Progress



Problem:

Land acquisition is one of the most critical and **time-sensitive phases** of infrastructure development. Delays in acquiring land significantly impact the execution of national and state-level projects. The causes of land acquisition delays are multifaceted, including prolonged administrative approvals, **legal disputes**, delayed **compensation disbursement**, incomplete documentation, pending notifications, **land ownership conflicts**, rehabilitation and resettlement challenges, and inter-departmental coordination issues.

Solution:

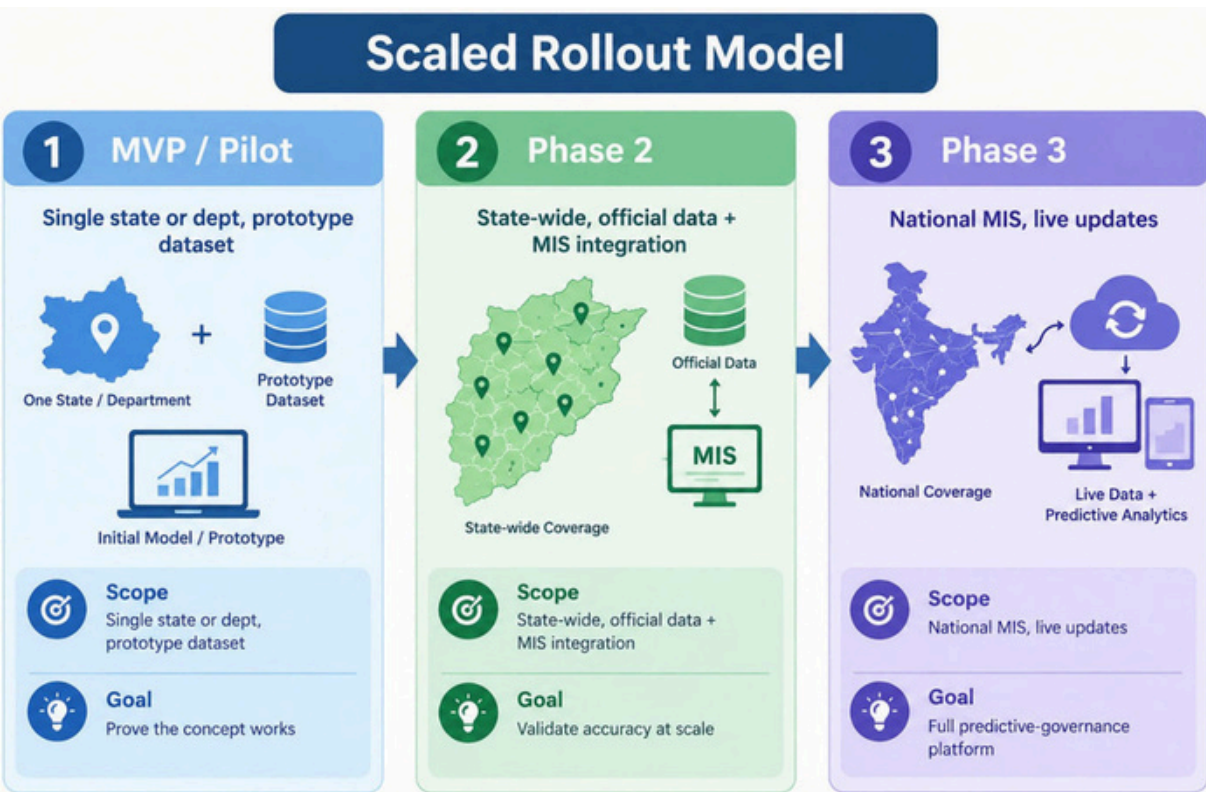
The system uses machine learning (Random Forest, XGBoost) on validated project data to **predict delay probability** and **delay duration** at each land-acquisition stage. **Explainable AI (SHAP)** reveals the key factors driving each risk score, while a What-If Simulation Engine lets officials test interventions and see the projected risk reduction instantly. A **GIS-based dashboard** with automated alerts **visualizes high-risk projects** for quick action, and a recommendation engine suggests corrective steps , **keeping humans in control of all final decisions**. This shifts governance from **reactive** to **predictive** to **preventive**.

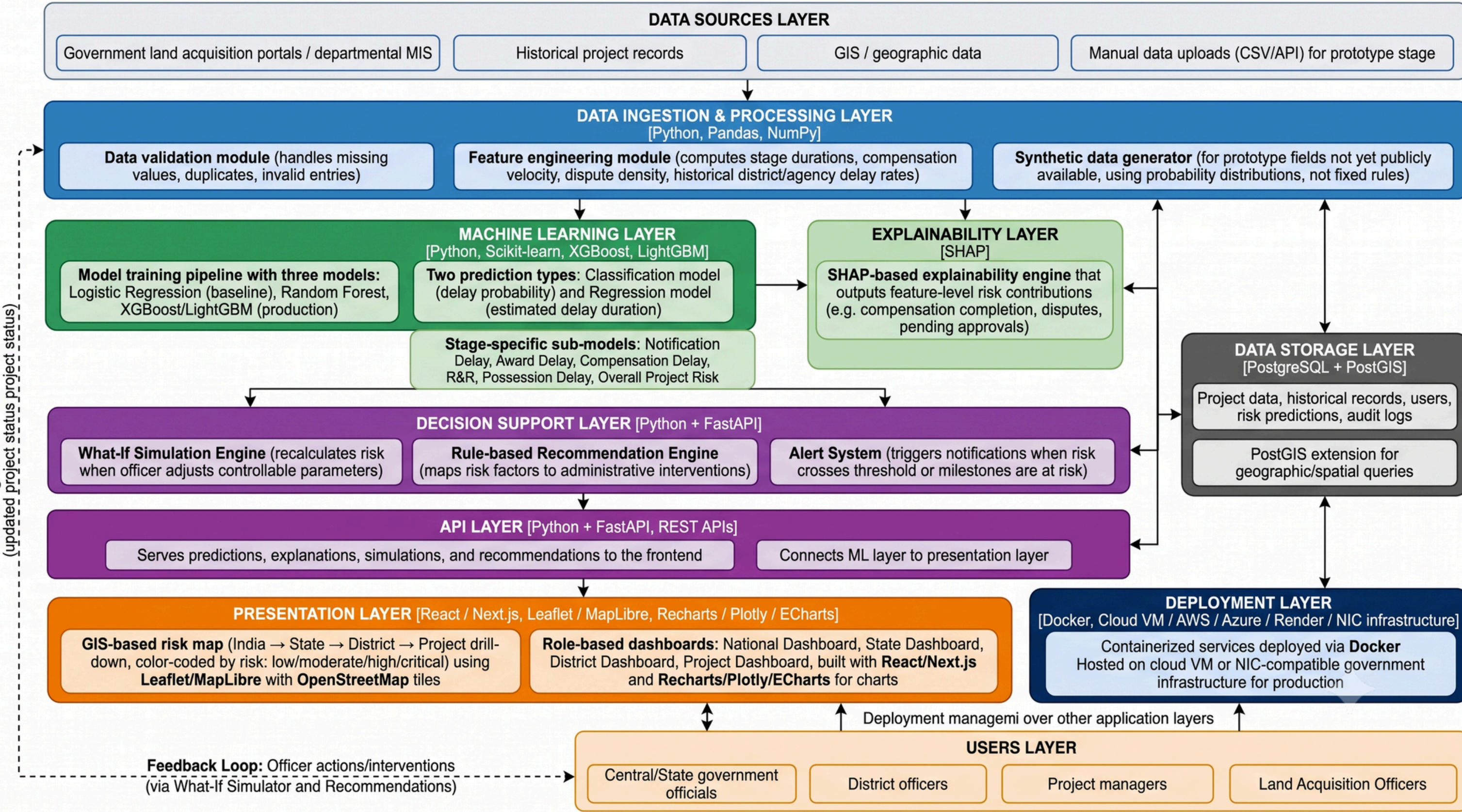
How It Addresses the Problem

 Predictive Warning System Forecasts delays and durations across milestones using ML classification and regression models (Logistic Regression, Random Forest, XGBoost).	 Explainable AI (SHAP) Identifies exact root causes of delay risk , such as low compensation completion, pending approvals, unresolved disputes , or lagging R&R.	 What-If Simulations Enables authorities to model interventions in real time (e.g., accelerating compensation or clearing disputes) to quantify risk reduction before acting.	 Stage & GIS Intelligence Tracks granular risk across all lifecycle stages (Notification to Possession) and maps high-risk hotspots for district-level oversight.	 Actionable Recommendations Delivers prioritized, rule-based operational fixes, such as scheduling compensation camps , escalating disputes, reallocating field resources .
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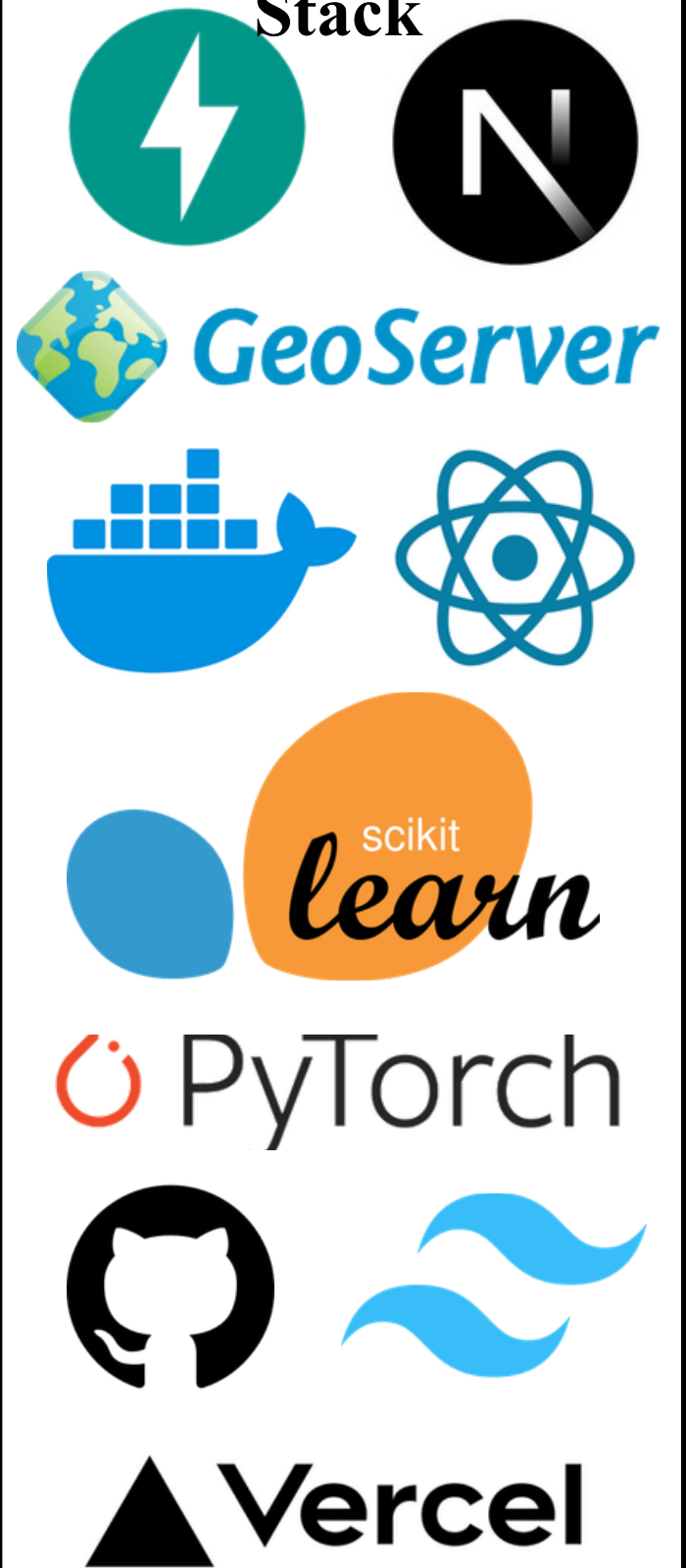
Innovation and Uniqueness

- **Stage-wise, explainable prediction:** forecasts delay risk separately for each stage (Notification, Compensation, R&R, etc.) and uses SHAP to show why, not just a single black-box score.
- **What-If Simulator :** lets officers test interventions and see risk recalculate live, turning the tool from a passive **dashboard into an active decision-making engine**.





Technology Stack



Technical Feasibility

- Built on proven models **Logistic Regression, Random Forest, XGBoost, paired with SHAP** for explainability and GIS for mapping.
- Uses a mix of public and synthetic data during the prototype phase, so development doesn't get stuck waiting on restricted government datasets.
- Pipeline is modular, so **each part can be improved or swapped** without breaking the rest.

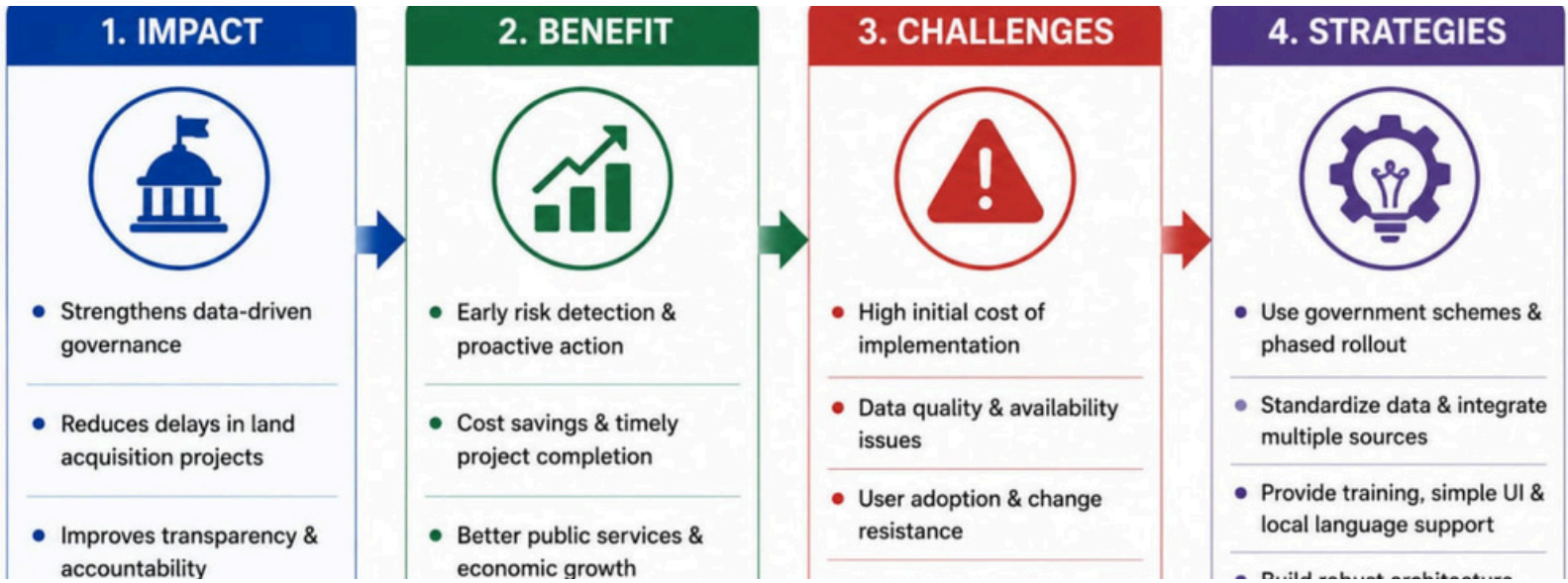
Operational Feasibility

- Connects to existing land-acquisition portals and **MIS via API/CSV**, no need to rip out current infrastructure.
- Role-based dashboards (Central, State, District, PM, LAO) are **simple enough for non-technical officials** to actually use.
- **AI only recommends; officers still decide.** That keeps people comfortable with it instead of resisant to it.

Economic Feasibility

- Sits on top of existing systems rather than requiring new infrastructure, so deployment **cost stays low.**
- **Early risk detection** saves money down the line fewer cost overruns, less litigation, less lost to delay.
- Directs limited government resources toward **the highest-risk projects** first, instead of spreading them thin.

Challenges	Strategies
Official land-acquisition data is restricted.	Use hybrid public + synthetic data now; request anonymized data for Phase 2
Government records are often messy or incomplete	Dedicated validation layer catches missing values, duplicates, outliers
Historically slow districts could get unfairly flagged	Regular fairness checks ; geography never decides alone, human stays in the loop
Officials may distrust AI-driven predictions	SHAP explains every prediction; AI recommends, officers always decide



SOCIAL IMPACT

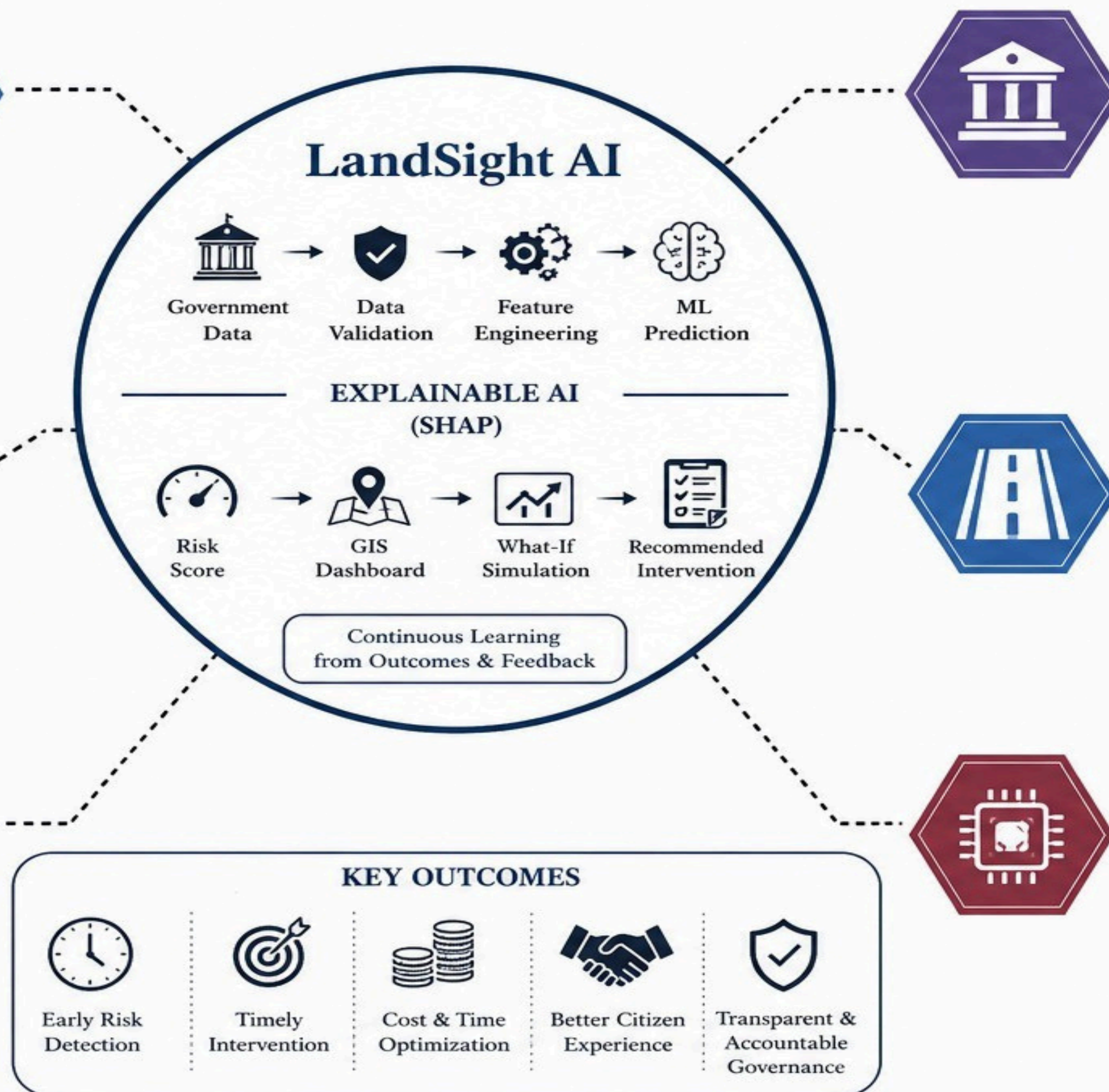
- Faster identification of land acquisition bottlenecks helps reduce project delays
- Timely compensation & R&R monitoring improves citizen trust and satisfaction
- Greater transparency reduces disputes, grievances and social unrest
- Strengthens collaboration between departments, citizens and stakeholders
- Better rehabilitation tracking ensures better support for affected families

ECONOMIC IMPACT

- Early risk detection helps prevent delay related cost escalations
- Optimized resource allocation across high-risk projects and critical stages
- Reduces idle time and cost overruns in infrastructure execution
- Supports timely land availability, enabling faster construction and earlier economic returns
- Improves overall efficiency of public spending and ROI on infrastructure

ENVIRONMENTAL IMPACT

- Faster project execution reduces extended construction time and associated emissions
- GIS-based monitoring minimizes ecological disruption and encroachment
- Supports sustainable land-use planning and compliance with environmental norms
- Digital monitoring reduces paper-based processes and administrative overhead
- Promotes responsible development of infrastructure with lower environmental footprint



GOVERNANCE IMPACT

- Data-driven governance with predictive insights and early warnings
- Evidence-based decision making rather than reactive interventions
- SHAP-based explanations increase transparency and accountability
- Prioritized intervention helps officers focus on critical projects and stages
- Strengthens monitoring, evaluation and inter-department coordination

INFRASTRUCTURE IMPACT

- Ensures timely Right-of-Way (RoW) availability for linear & industrial projects
- Helps prevent project delays due to land acquisition bottlenecks
- Improves project planning with accurate risk visibility and simulation
- Supports smoother implementation of roads, railways, power, industrial and urban infrastructure
- Reduces probability of stalled or abandoned projects

TECHNOLOGY IMPACT

- AI/ML models provide accurate risk prediction and stage-wise intelligence
- GIS visualization enables spatial analysis and hotspot identification
- What-if simulation supports proactive planning and intervention evaluation
- Scalable, secure and interoperable digital platform
- Continuous learning improves model accuracy and system intelligence

Delay Analysis of Infrastructure Construction Projects in India Author: L. Pinky Devi, Sindhu 2025	<u>LINK</u>
A Study of Land Acquisition and Compensation Process for National Highway in Maharashtra Authors: Renuka R. Purohit, Rahul S. Chaudhari, Ankit Saraswat 2023	<u>LINK</u>
Streamlining the M&E of Land Acquisition and R&R in Large Infrastructure Projects Using IT Author: Ramakrishna Nallathiga NICMAR, 2013	<u>LINK</u>
Mega Projects in India: Environmental and Land Acquisition Issues in the Road Sector Authors: G. Raghuram, Samantha Bastian, Satyam Shivam Sundaram (IIM AHMEDABAD)	<u>LINK</u>
Land Acquisition and Compensation in India: Mysteries of Valuation Author: Sattwick Dey Biswas	<u>LINK</u>
Money, Land or Self-employment? Understanding Preference Heterogeneity in Landowners' Choices for Compensation under Land Acquisition in India Authors: Vikram Patil, Ranjan Ghosh, Vinish Kathuria, Katharine N. Farrell	<u>LINK</u>